SIMATS ENGINEERING
ASSESSMENT TOOLS I

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA13O1**

COURSE OUTCOME COVERED

CO3: *Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems (BL:3)*

Assessment Tool Weightage: 5%

QUESTIONS: 15 **Total Marks:30**
Duration : 45 Minutes

MCQ

Code of Conduct:

I Deepak V-192411021 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

Q1. Which of the following represents the language of all strings over {a, b} ending with 'a'?

A) (a+b)*a
B) a(a+b)*
C) (a+b)a*
D) a*(a+b)

A

Q2. The regular expression (ab)* generates:

A) a, b, ab
B) ab, abab, ababab
C) aabb, abab
D) ba, abba

B

Q3. Which regular expression represents all strings containing at least one 'a'?

A) b*
B) (a+b)*a(a+b)*
C) a*
D) (ab)*

B

Q4. The language of the regular expression a*b* is:

A) All strings with equal a's and b's
B) All strings with a's followed by b's
C) Strings with alternating a and b
D) Only a's and b's separately

B

Q5. Which of the following is NOT a regular expression?

A) a*
B) (a+b)*
C) $a^n b^n$
D) ab+a

C

Q6. Constructing Regular Expressions

Q6. Which regular expression denotes strings with exactly two 'a's?

A) (a+b)*aa(a+b)*
B) a*aa*
C) (b*ab*ab*)
D) Both A and C

C

Q7. Language of strings starting with 'a' and ending with 'b':
A) a(a+b)*b
B) (a+b)*ab
C) ab(a+b)*
D) a*b*

A

Q8. Which expression represents all strings over {0,1} with even number of 0's?
A) (1*01*01*)*
B) (01)*
C) 0*1*
D) (10)*

A

Q9. Pumping Lemma
Q9. Pumping Lemma is used to:
A) Prove a language is regular
B) Prove a language is not regular
C) Construct DFA
D) Simplify expressions

∴ B

Q10. In pumping lemma, the string is divided into:
A) x, y
B) x, y, z
C) a, b, c
D) p, q, r

B

Q11. Condition for pumping lemma:
A) $|xy| \le p$ and $|y| > 0$
B) $|x| \ge p$
C) $|y| = 0$
D) $|z| \le p$

A

Q12. Which language is NOT regular?
A) a*b*
B) (ab)*
C) $a^n b^n$
D) (a+b)*

C

Q13. Closure Properties
Q13. Regular languages are closed under:
A) Union
B) Intersection
C) Complement

D) All the Above

D) All of the above

Q14. If L_1 and L_2 are regular, then $L_1 \cap L_2$ is:
A) Not regular
B) Regular
C) Context-free
D) Undefined

B

Q15. Which operation is NOT a closure property of regular languages?
A) Union
B) Concatenation
C) Kleene Star
D) Infinite memory

D

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (25 Marks)	Level 3 – Proficient (20-24 Marks)	Level 2 – Developing (10–15 Marks)	Level 1 – Beginning (5 -10 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

92/100